

Supplementary Materials: Edge-Level Discrete Curvature Decomposes Bridge Structure Across Disease Networks

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S1 Full MR Results

Table S1 reports IVW, MR-Egger, and weighted median estimates for all 49 completed cross-disorder MR pairs. Seven ASD-as-exposure pairs are omitted (insufficient instruments). PTSD-as-exposure betas are inflated by Z-to-beta conversion and should not be compared in magnitude to OR-scale GWAS. Egger intercept p -values test for directional pleiotropy; values < 0.05 suggest the IVW estimate may be biased. I^2 values are available only for pairs with ≥ 25 instruments.

S2 Significant Edge-Level Curvature z -Scores

Table S2 lists the 32 edges with $|z| > 1.96$ against the degree-preserving null (200 permutations) in the expanded 84-entry graph. All four bottleneck edges involve depression or bipolar disorder bridging cross-domain mechanisms. The 28 redundant edges include domain-specific clusters (autoimmune, thyroid, microbiome) and encapsulated claims.

S3 Mechanism Catalog

Table S3 lists all 47 entries in the literature-derived catalog from which the mechanism network is constructed. Verdict tiers range from Disconfirmed (lowest) to Validated (highest). Shared mechanism nodes link claims across disorder families; entries with no shared nodes connect only to their disorder family.

S4 Robustness Details

Table S4 reports catalog-perturbation survival rates for the expanded 84-entry graph (100 subsamples of 71/84 entries, 20 null permutations per subsample).

S5 Expanded Catalog: Cross-Domain Entries

Table S5 lists the 37 cross-domain MR entries added to the core 47-entry catalog. These entries span sleep-mood, metabolic, personality, substance use, inflammation, microbiome, thyroid, autoimmune, and neuroimaging domains.

Table S1: Full cross-disorder MR results (49 pairs). k : number of independent instruments after LD clumping. [†]Egger intercept $p < 0.05$, indicating possible directional pleiotropy; IVW estimate should be interpreted cautiously. PTSD-as-exposure betas are inflated by Z-to-beta conversion.

Pair	k	IVW β	95% CI	Egger β	Egger int. p	WM β	$I^2\%$
ADHD $\rightarrow$ ANX	25	0.167	[0.135, 0.199]	–	0.303	0.183	–
ADHD $\rightarrow$ ASD	25	0.482	[0.391, 0.573]	–	0.080	0.469	–
ADHD $\rightarrow$ BIP	24	0.182	[0.131, 0.232]	–	0.732	0.128	–
ADHD $\rightarrow$ MDD	22	0.130	[0.110, 0.150]	–	0.164	0.119	–
ADHD $\rightarrow$ PTSD	24	0.084	[0.073, 0.096]	–	0.574	0.079	–
ADHD $\rightarrow$ SCZ	25	0.303	[0.246, 0.359]	–	0.174	0.259	–
ADHD $\rightarrow$ SUD	8	0.064	[0.038, 0.089]	–	0.597	0.071	–
ANX $\rightarrow$ ADHD	47	0.535	[0.446, 0.623]	–	0.234	0.397	–
ANX $\rightarrow$ ASD	49	0.402	[0.273, 0.530]	–	0.682	0.402	–
ANX $\rightarrow$ BIP	46	0.588	[0.515, 0.660]	–	0.637	0.531	–
ANX $\rightarrow$ MDD	43	0.509	[0.480, 0.537]	–	0.329	0.517	–
ANX $\rightarrow$ PTSD	46	0.251	[0.234, 0.268]	–	0.457	0.245	–
ANX $\rightarrow$ SCZ	48	0.773	[0.692, 0.855]	–	0.513	0.641	–
ANX $\rightarrow$ SUD	14	0.135	[0.096, 0.175]	–	0.738	0.145	–
BIP $\rightarrow$ ADHD	58	0.136	[0.085, 0.187]	–	0.707	0.126	–
BIP $\rightarrow$ ANX	55	0.176	[0.148, 0.203]	–	0.886	0.183	–
BIP $\rightarrow$ ASD	58	0.219	[0.145, 0.294]	–	0.029	0.208	–
BIP $\rightarrow$ MDD	47	0.169	[0.152, 0.186]	–	0.084	0.167	–
BIP $\rightarrow$ PTSD	53	0.072	[0.062, 0.082]	–	0.955	0.064	–
BIP $\rightarrow$ SCZ	60	0.634	[0.588, 0.680]	–	0.532	0.613	–
BIP $\rightarrow$ SUD	14	0.050	[0.026, 0.074]	–	0.343	0.043	–
MDD $\rightarrow$ ADHD	124	0.959	[0.875, 1.043]	0.960	0.928	0.971	58
MDD $\rightarrow$ ANX	128	0.862	[0.819, 0.905]	0.857	0.419	0.874	47
MDD $\rightarrow$ ASD	126	0.572	[0.448, 0.695]	0.587	0.312	0.472	34
MDD $\rightarrow$ BIP [†]	126	0.774	[0.708, 0.841]	0.743	0.008	0.803	69
MDD $\rightarrow$ PTSD	125	0.339	[0.323, 0.355]	0.337	0.467	0.334	46
MDD $\rightarrow$ SCZ	128	0.909	[0.833, 0.985]	0.897	0.520	0.825	85
MDD $\rightarrow$ SUD	59	0.240	[0.210, 0.269]	0.243	0.568	0.252	68
PTSD $\rightarrow$ ADHD	53	2.036	[1.816, 2.256]	–	0.950	2.223	–
PTSD $\rightarrow$ ANX	51	1.579	[1.462, 1.696]	–	0.710	1.624	–
PTSD $\rightarrow$ ASD	53	1.158	[0.837, 1.480]	–	0.367	1.111	–
PTSD $\rightarrow$ BIP	51	1.359	[1.181, 1.536]	–	0.582	1.623	–
PTSD $\rightarrow$ MDD	44	1.394	[1.320, 1.468]	–	0.447	1.318	–
PTSD $\rightarrow$ SCZ	54	1.706	[1.506, 1.905]	–	0.564	1.810	–
PTSD $\rightarrow$ SUD	25	0.455	[0.383, 0.527]	–	0.395	0.498	–
SCZ $\rightarrow$ ADHD	105	0.083	[0.054, 0.113]	–	0.858	0.083	–
SCZ $\rightarrow$ ANX	103	0.137	[0.122, 0.153]	–	0.674	0.128	–
SCZ $\rightarrow$ ASD	106	0.153	[0.109, 0.196]	–	0.981	0.137	–
SCZ $\rightarrow$ BIP	103	0.334	[0.310, 0.359]	–	0.115	0.326	–
SCZ $\rightarrow$ MDD	98	0.091	[0.082, 0.100]	–	0.754	0.097	–
SCZ $\rightarrow$ PTSD	102	0.050	[0.044, 0.055]	–	0.719	0.046	–
SCZ $\rightarrow$ SUD	55	0.051	[0.041, 0.061]	–	0.683	0.042	–
SUD $\rightarrow$ ADHD	15	0.963	[0.629, 1.296]	–	0.844	1.338	–

Table S2: Significant edge-level curvature z -scores (32 of 103 tested edges). Sorted by z -score.

Edge	κ_{obs}	z
<i>Bottleneck edges ($z < -1.96$)</i>		
Depression – convergence hub	−0.867	− 5.46
Depression – metabolic	−0.780	− 3.93
Bipolar – metabolic	−0.693	−3.33
Depression – circadian	−0.675	−2.62
<i>Top redundant edges ($z > +1.96$; 28 total)</i>		
Autoimmune cluster	+0.583	+ 9.40
FKBP5 – claim 004	+0.167	+5.09
ASD – claim 037-inst	+0.167	+4.15
HPA axis – claim 004	+0.104	+4.09
Thyroid-Metabolic – thyroid	+0.100	+3.60
Microbiome – gut-brain	+0.267	+3.43
Anxiety – claim 023	+0.000	+3.39
Autoimmune – claim 076	+0.250	+3.38
Sleep-Mood – circadian	+0.100	+3.26
Schizophrenia – claim 029	−0.015	+3.14

Table S3: Full mechanism catalog (47 entries). Verdict tiers: D = Disconfirmed, U = Under-determined, I = Inconclusive, CS = Causally Suggestive, MS = Mechanistically Supported, V = Validated.

ID	Family	Verdict	Shared mechanisms
001	Depression	D	–
003	Depression	D	Serotonin
004	Depression	D	HPA axis, FKBP5
006a	Depression	CS	IL-6
006b	Depression	D	IL-6
008	Depression	D	–
008b	Depression	MS	–
010	Depression	I	–
011	Depression	CS	NMDA/Glu, IL-6
012	Depression	U	NMDA/Glu
012a	Depression	MS	NMDA/Glu
013	Depression	U	Mitochondrial
014	Depression	CS	Circadian
015	Depression	CS	FKBP5, HPA axis
016	Depression	CS	–
017	PTSD	CS	FKBP5, HPA axis
018	PTSD	MS	Fear circuit
019	PTSD	CS	Noradrenergic
020	PTSD	U	HPA axis
021a	Anxiety	V	–
021b	Anxiety	D	–
022	Anxiety	MS	Fear circuit
023	Anxiety	D	Serotonin
024	Anxiety	CS	Noradrenergic
025	Schizophrenia	MS	Dopamine
025-orig	Schizophrenia	D	Dopamine
026	Schizophrenia	CS	NMDA/Glu
027	Schizophrenia	MS	Syn. pruning
028	Schizophrenia	CS	MIA/prenatal
029	Schizophrenia	CS	MIA/prenatal, IL-6
030	Bipolar	CS	Mitochondrial
031a	Bipolar	CS	Circadian
031b	Bipolar	D	Circadian
032	Bipolar	CS	–
033	Bipolar	I	–
034	OCD	MS	CSTC circuit
035	OCD	D	Serotonin
036	OCD	U	NMDA/Glu
037	ASD	CS	NMDA/Glu
037-inst	ASD	MS	NMDA/Glu, Syn. pruning
038	ASD	CS ⁴	MIA/prenatal, IL-6
039	ASD	MS	Syn. pruning
040	Addiction	MS	Dopamine

Table S4: Catalog perturbation survival rates (100 subsamples of 71/84 entries). Survival = fraction of iterations where the edge reaches $|z| > 1.96$. Only edges with $> 50\%$ survival shown.

Edge	Survival	Present
<i>Bottleneck edges</i>		
Depression–convergence hub	100.0%	99
Depression–metabolic	100.0%	96
Bipolar–metabolic	81.8%	22
Depression–circadian	76.3%	97
<i>Redundant edges</i>		
Claim 011–Depression	78.1%	32
IL-6–Inflammation	71.4%	14
Claim 015–Depression	60.6%	33
Depression–IL-6/inflammation	53.0%	100

Table S5: Cross-domain MR catalog entries (37 entries). Verdict abbreviations as in Table S3.

ID	Family	Verdict	Shared mechanisms
045–048	Sleep-Mood	CS/U	Circadian
049	Eating	MS	Metabolic
050	Eating-Mood	CS	Metabolic
051	Personality	MS	Depression convergence
052–053	Personality-Social, Social	CS	Depression convergence
054	Substance-Neurodev	MS	Dopamine
055	Substance-Mood	CS	Depression convergence
056	Substance	U	–
057	Neurodev-Mood	CS	Depression convergence
058	Stress-Mood	CS	FKBP5
059–061	Inflammation	CS/D	IL-6/inflammation
062–063	Microbiome	U	Microbiome-gut-brain
064	Mediator	MS	IL-6/inflammation
065	Microbiome-Neuro	CS	Microbiome-gut-brain
066–068	Thyroid-Metabolic	CS/U	Metabolic
069–072	Thyroid-Metabolic	CS/D	Thyroid
073	Thyroid-Metabolic	CS	Metabolic
074–078	Autoimmune	CS/MS/U	Autoimmune
079–081	Neuroimaging-endo	CS	Neuroimaging